YOUR MIX SUCKS

FOCUS-ON-VOCALS EXCERPT

TAKEN FROM
THE COMPLETE
MIX METHODOLOGY
FROM DAW PREPARATION
TO DELIVERY

www.mixedbymarcmozart.com

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FOREWORD TO THIS EXCERPT

This PDF is an excerpt from the full YOUR MIX SUCKS - Waves Edition. The first edition of the full book was written in 2014. It was followed by mix templates and channel strip libraries for your DAW that contain all the principles, signal chains and methods used in this book (MIX TEMPLE PRO).

Before we start, I'd like to invite you to join me on Facebook, Instagram, YouTube and LinkedIn. I am using all aforementioned channels as well as my podcasts (available in English and German language) to deliver a lot of free audio mixing and social-media marketing know-how for producers, DJs, singer/songwriters and audio engineers. Also, all legitimate buyers of the full e-book receive a year of free mix feedback from me - you just e-mail me your track to feedback@mixedbymarcmozart.com and I'll come up with some quick suggestions for improvement. Of course, you can use that same e-mail for all sorts of feedback, questions and whatever you need from me.

Marc Mozart, November 2020

INTRODUCTION

Strategically planned as this book might look, it started out by many of my friends asking me the following or a similar question almost every day:

> *"Hey Marc, you know I've been doing this for a while. I'm really happy how my songs turn out, and I've come a long way as a producer. But to be totally honest: my mix sucks. Do you have any basic advice how I can improve my mixing?"*

The content of this book is the full version of what I would say on such occasions. Many people still reckon that mixing is "turning knobs 'til booty shakes" or "some levels, compression and EQ, until it all sounds right".

In this book, we explore things a little more systematically. We talk about cause and effect, why things have to be done in the right order, and what path to work your way through to a great mix. The solution lies in the process and the order in which you sort out things.

If you don't have a methodology, your mix can suffer from a severe "domino effect". Which of course is an analogy for one bad decision triggering a series of follow-up effects, and while you're working towards finishing your mix, what started as a small problem will accumulate and get worse. For example: A bad listening environment will inevitably lead to wrong mixing decisions that you'll be busy correcting all the way through - instead of really focusing on the actual creative elements of mixing. You're getting all frustrated about the outcome of your mix, while in reality, you had no chance getting it right under the given circumstances.

Later on, when you check your mix in the car, in a club, or even at the mastering studio, it's too late. I fully trust that you have the ears to hearing something is wrong with your mix, but your problem has gotten out of hand with a series of follow up decisions based on the wrong assumptions. A good methodology helps to avoid domino effects by dealing with fundamental problems first. It's also supporting your creativity and intuition, because all of a sudden you're not putting out fires all the time, you're making real progress. To further explore this I'll give you three examples.

1. FIRST THINGS FIRST:

Look into your monitoring

Big decisions require small speakers, small decisions require big speakers. Especially when your room is untreated, there's no point in using huge monitors with massive low end. Instead, mix through a "smaller window" and make decisions as if you were walking by.
We'll deal with this topic from the very beginning and there's a major "hack" that many of the world's greatest engineers are using. Hint: is has to do with your mom's kitchen ;-)

Treat your room

Less than $ 300 in self-made room treatment can make a huge difference for your mix room.

Know the musical context of your low end

Look at kick and bass in a musical context, and increase your chances of organically sorting out your low end.
If the fundamental frequency of your kick corresponds with the song's key, and the bass-notes aren't out of hand dynamically, things are a lot more likely coming together, as opposed to doing some random EQing in the low end.

2. THE "BOTTOM-UP APPROACH"

Because low frequencies (= bottom) have a lot more energy than mids or highs, we always take care of them first.

The "bottom-up approach" in EQing

Get the lower frequencies of each sound right before you shape mids and highs. Look at each sound consisting of a tone and a noise component.

After removing unwanted low end rumble, make sure to get the tone right. The tone is based on the 1st and 2nd harmonics of each sound – make sure it's round, well-balanced and not overly resonant.

Once you got the tone right, look at the balance of tone and noise in each sound. This can be as simple as a high shelf EQ above 3K, adding frequencies if you want more sizzle in a synth, more fret-noises in a guitar, or more click in a kick drum.

The "bottom-up approach" in compression

The meters in your DAW can fool you. Spikes on peak meters are often caused by high frequencies, and the first intuition leads musicians to apply compression or limiting to get a more controlled sound by reducing those. Well, no – reducing or removing peaks using a super-fast compressor or brick wall limiter at this stage won't give you a very musical result.

In reality, the loudness (= RMS or LUFS level) of your signal is a lot more affected by the lower frequencies in your signal. Unless you are dealing with drums or percussion, the first compressor in your plugin or effect-chain needs to be a slower, more musical compressor. More musical meaning with a slow attack time, a low compression ratio, high threshold, and many times, a compressor that adds tone via harmonic distortion which makes your low end more perceptible on smaller speakers.

3. INSTALLING HANDLES

The notion that you can just set a sound and forget it while you're mixing, is a misconception.
Take a kick drum for example, you will never be able to determine the final amount of low end in the kick out of context with the bass, and of course your mix bus and mastering-chain.

On some mixes the kick already comes with a lot of low end, on others it's slim but well-balanced with the bass, and you dial in the final amount of low end with a Pultec on your mix bus.

Let's not forget why a mastering-guru like Brian Gardner has the nickname "Big Bass" – if your mix is well-balanced and dynamic, your mastering engineer has a much better chance to give you the maximum low end energy as opposed to you boosting your kick at the start of your mix, and slamming 0dB on the digital meters right away.

"But how do you control these things throughout the different stages of your mix?"

Here's where the "handles" come into play, which could also be called "control points". Let's stay with the example of the kick.

The basic handles would be:

- A roll-off for unwanted subsonic energy (between 0Hz and 60Hz)
- A bell curve EQ on your exact fundamental frequency of your kick (between 40Hz and 70Hz) aka "1st harmonic": this is what the fundamental frequency of a sound is called, for an exact physical explanation, here's a nice Wikipedia article
- A bell curve EQ on the 2nd harmonic aka pressure point of your kick (an octave up from the fundamental = between 80Hz and 140Hz)
- A shelf EQ for anything higher than 300Hz

- A transient shaper/designer or envelope to control the level of the initial transient aka "click" of your kick, and the length of the "tone".

Installing handles doesn't mean that you boost or cut anything right away, but identifying these frequencies and setting them will allow you to have full control over your Kick throughout the entire mixing process and to deal with the balance if and when that decision comes up. Towards the end of your mix, you might find that you need to cut a bit of the fundamental to make space for your bass, or to boost the second harmonic to give it more punch.

If you use a final mix bus EQ to give your entire mix more "oomph", you will very likely need to reduce the fundamental tone, to have more leeway for the bass-boost. There are a lot of variables, and the handles are giving you access to those parameters that matter and will help you deal with every issue.

This is one reason why I'm proposing to work with a mix template that has your routing, VCA-Groups and most commonly used FX busses already in place. Beginners often believe that a mix template does the listening and mixing for them while a better analogy would be that the studio and gear is wired and powered so you can start making music instead of building a studio. A blank DAW session is a bit like an empty studio with all the gear still packed in boxes.

The concept of handles applies to any sound in your mix, but the type of parameters are obviously different depending on the source.

Just to give you another example, the following are the most useful handles on a lead vocal.

Typical lead vocal handles are:

- Again, a roll-off for unwanted subsonic energy (between 0Hz and 140Hz)
- Sibilance: a De-Esser to reduce it if necessary combined with a shelf EQ to fine-tune the amount of overall treble you need in the vocal
- Weight and warmth: a wide bell EQ around the fundamental notes of the lead vocal
- Presence: various EQs around the mids, can range between 300Hz – 8kHz
- Attitude: a compressor that keeps the lead vocal upfront relative to the other signals

The point of installing handles on your source material is that they will be there when you need them. At any stage of your mixing process you have the tools at hand to re-balance all tonal and dynamic aspects of your mix. These are often subtle changes, but the reason we need access to them all the time, not just when an issue comes up, is to avoid what this introduction was about: the "Domino Effect", or how one bad decision triggers a series of follow up problems.

Don't worry, if that introduction was a little too much information at once! I just wanted to give you a rough idea around the philosophy behind this book. The book will go through all of that in a logical order, step by step.

Enjoy :-)!

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CHAPTER 06

IT'S ALL ABOUT THE VOCALS

IT'S ALL ABOUT THE VOCALS

Once we have created a frame for our mix as described in chapter 5, it's time to focus on the vocals - in many cases the most important instrument in the mix.

The human voice is the most complex instrument we have to deal with, and in the following, I'll introduce you to the important aspects of dealing with vocals in a mix.

VOCAL RECORDING - PREPARATION

I know this is not directly related to mixing, but a great vocal performance and recording is the best foundation for a quality vocal in the mix - so if you're in charge of vocal production, here are some ideas for you.
Getting this right can save you a lot of work in the mix!

1. GET THE SINGER TO DO A VOCAL WARM-UP

There are some great vocal warm-up programs out there. Just do a search on YouTube for "vocal warm up" and you'll find lots of options. The warm up should take at least 30 minutes, a full hour is even better. A good warm up has a slow build, and consists of a big variety of notes, expressions and phrases.

Trust me, it's worth it and will save you hours in vocal recording. It's basically like sports - no professional athlete would ever skip preparation and warm up! My favorite vocal warm up is by Stevie Lange - on her website you can find female and male versions.

2. GET THE SINGER TO DRINK LOTS OF STILL WATER

In addition to the warm up, the singer should drink lots of still water before and also during the session.

3. PRINT OUT THE LYRICS

This will be helpful for both the producer and the singer. Type down the lyrics of the song and print 2 copies, one for you, one for the singer. Make sure the font is large enough so it's easy to read. Print it on two pages if it doesn't fit on one.

Have a music stand and a working pen ready for the singer and also yourself, so you can both make notes. Place the music stand and note sheet in a way that it doesn't reflect back into the microphone - the right angle helps wonders!

4. RECORD A COUPLE OF RUNS THROUGH THE ENTIRE SONG

I know – recording vocals in sections is very popular, but it's still a good idea to record the entire song from start to finish at the beginning of the session. Mainly because it gives you a better idea of the macro dynamics of the song.

When recording a song in sections, you can easily overpower the first verse, then realize the singer can't top the power in the chorus. So make sure to define the tone and power of each section during these run-throughs.

5. RECORD THE SONG IN SECTIONS

If the complete runs went very well, this is just a backup, but the reality is that most vocals are recorded in sections. Loop a song part (e.g. verse or chorus), make sure to have 2 - 4 bars pre-roll of just music between every take and just keep recording.

You will probably see the performance improve with every new take until at one point the quality drops which is when you should stop and go to the next section. Once you went through all the sections, you should start at the first section again because there was likely a development in quality, and this second session is usually a huge improvement over the first one.

6. HOW MANY TAKES?

The number of takes you want to capture varies - there is no rule to that. I've worked with professional session singers in New York that were so well prepared that 3 takes was all I've ever needed. In the 2000s, I've worked with a lot of young and inexperienced casting TV show artists (Idol, X-Factor, The Voice), where I would record a minimum of 10 - 20 takes of each section. You have to decide it from case to case, but keep in mind that sometimes, this one vocal session is your only chance to get it right, and better be safe than sorry!

VOCAL RECORDING - WHAT TO AVOID

Now on to the more technical aspects of vocal recording - here are some of the biggest issues I see:

1. TOO MUCH TREBLE AND "ESSES"

Vocal sounds are a matter of trends. For years, the Sony C800G tube microphone has been very popular due to its emphasis and fine resolution on higher frequencies. Without a doubt, one of the best mics ever made, but at the same time, depending on the type of vocal, there might be excessive sibilance on the vocal-tracks.

A mic that emphasizes the highs is not the main source of trouble here - its excessive sibilance pushed into a tracking compressor that creates the problem. For this reason, I don't recommend a tracking compressor when trying to achieve a vocal sound with crisp highs.

Along with more organic productions, recent years brought back more natural, organic-sounding vocals, with less emphasis on the highs.

Another positive development for music producers is that the market for affordable studio mics is growing and changing. Classic microphone designs are getting very affordable, costing a fraction of the price of 10 years ago, and the latest trend of DSP-modeled microphones is already giving us a glimpse into the future.

2. NO POP SHIELD

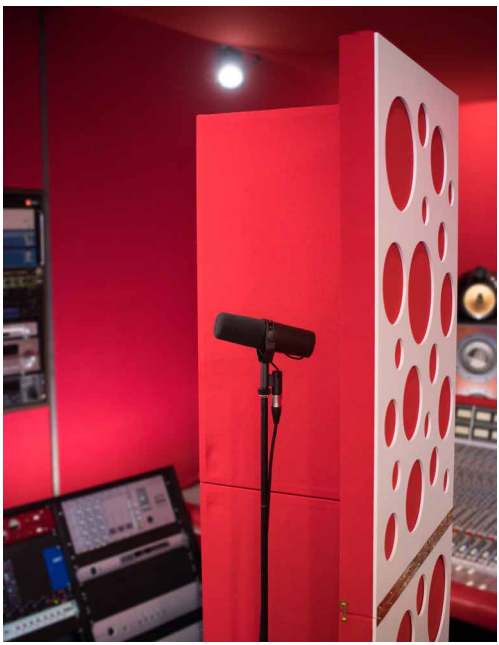
If you don't use a pop shield in front of your vocal mic, chances are you will literally have dropouts in your audio signal. Not every pop shield sounds the same - it's definitely worth having a few different ones at hand and listen to the difference.

3. RECORDING IN A VOCAL BOOTH

Most small home-made vocal booths are not properly absorbing (= reducing reverb-time of the room) the sound across the entire frequency spectrum. All they do is dampening the highs, leaving you not only with a boomy sound, but also with comb-filtering that results in resonances across the mids.

Vocals sound better recorded in larger rooms, and it is a lot easier to deal with a few more reflections from a large room than removing resonances from comb-filtering.

Of course, an ideal environment is spacey (= lots of space between the singer and the next hard wall) and absorbing at the same time. If you have an acoustically treated control room, try recording the vocals there. Movable acoustic panels (aka GOBOs) can be very helpful as well. I usually record vocals in my large control room while putting the GIK Acoustic PIB (Portable Isolation Booth) around the back of the microphone.



Our portable acoustic booth by GIK Acoustic

4. USING A TRACKING COMPRESSOR

Again, if in doubt, don't use a tracking compressor – it's a habit from the analog tape tracking age, where high recording levels where the treat against the ever existing noise floor of tape. In a digital 24bit world, set your mic-pre and tracking levels carefully, and make sure you have plenty of headroom even during the loudest vocal parts of the song.

A high quality mic pre and A/D converter can be a good investment, but to be fair, even lower priced USB audio interfaces provide lots of dynamic in 2019.

VOCAL TUNING

For me, tuning vocals is part of fixing the performance and I'm not a big fan of it as part of the mixing process. It can take a lot of time and attention away from getting the mix right in a macro, but since it's technically possible, it's part of our repertoire of tools.



The Waves Tune Real-Time Vocal Tuning Plugin



The Waves Tune: When you need total control

Here are some basics on using tuning plugins in a mix:

1. Basic Setup

Once more, you need to know the key of the song you're mixing and the exact notes covered by the vocals. Every tuning plugin has a setting for the song-key. Setting the song-key to "chromatic" works in theory, but can create notes that are completely wrong.

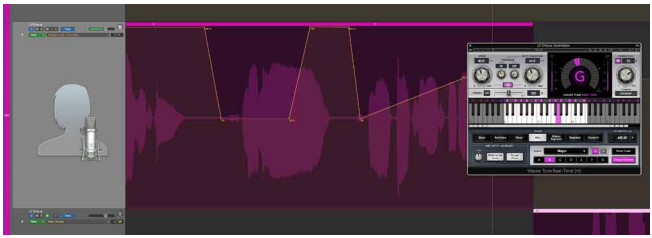
Also, it's not enough to set the key to C minor, when both B and Bb are possible notes depending on the context of the chord progression. Then, there's songs that modulate between different keys, in that case it's good to have separate vocal tracks for these sections. Once you have set the correct key, the setup is straight forward for most pop-songs or similar genres.

2. Automating Speed / Response Time

Vocal tuning plugins always have a speed-setting that determines the speed in which a note will be tuned. When it's set fast, you get that classic FX-sound, but when set to a slower value, the tuning plugin can be

very subtle as the plugin takes a long time to "bend" the tone to the target note, just like a good singer would do naturally. There's definitely a setting with a long speed time that makes vocal tuning completely unnoticeable. The next step is to draw the speed in the automation of your DAW.

If there's a note or phrase that needs tuning, you just automate the speed to a faster value. The automation curve you're using to go from slow to fast is the factor for how much you hear side-effects of tuning.



Automation curve on the Waves Tune Real-Time Vocal Tuning Plugin

I am of course assuming that you want the vocal to sound natural, with no side-effect. When using tuning plugins, avoid having all doubles of the same vocal-line tuned. Especially on a backing vocal, if you have two doubled takes, only tune one of them. If you have more than two, only tune half of them, and between the tuned ones, fine-tune them against each other (for example: left BV at -5 cent, right BV at +5 cent).

3. Vocal Comping

Another trick that needs to happen at production stage is having the tuning plugin in your monitor chain while comping the vocal takes from your recording session. That way, you can choose to not select takes that have noticeable side-effects /artefacts from the tuning plugin.

BUILDING THE LEAD VOCAL-CHAIN

All of the following, please do at a low listening level. I tend to use my small portable stereo-speakers (aka "kitchen radio") for that and start by leveling the untreated lead vocal so that it sits a little low in the track, just loud enough so you can understand the lyrics and follow the melody.

Let me make an important statement – and I'll say this a few more times in the book: keep going back between these different building blocks for fine-tuning! Just as an example, you will need to re-adjust the sibilance once you've added the final "attitude"-compressor. That goes for each building block. Go back and forth between the building blocks and re-adjust.

Also, watch your gain staging - make sure the input level at each building block is similar to the output level. This chain has 3 dynamics processors with EQs in between, they are kind of "sharing" the workload of making the vocal sit tight in the mix. None of these building blocks need to do the "heavy lifting" on their own, each has their own role, and less is more.

Bypassing one or more building blocks is always an option. Many times you don't need all of them! Before I go through the chain, I'd like to again stress the importance of sending the correct level into the plugin chain. Start with an average (RMS)-level of -18dBFS here (that's the scale in your DAW). If your recording is louder than that, lower it using a gain plugin.

1. CONTINUITY

Starting with the untreated, "naked" lead vocal-track, I want you to first understand the concept of having two types of volume automation on a vocal: before the processing chain, and after it.

This first one evens out the performance of the singer. You could say it creates the continuity of levels in the vocal that you wished the singer had delivered in the first place. This is something that overlaps and goes back to mix preparation, and there is a fair chance that this is already in a good place. To confirm this, run the mix, go through the song from start to end and find spots where phrases or syllables of the lead vocals drop in level, or jump out considerably. Correct these phrases in whole dB-values - in other words, don't go too much into detail, don't draw complex automation curves etc., this is just one small step to even out the lead vocal across the song. Waves has a plugin called Vocal Rider that can do the job. You can set a target-level for the vocals and depending on the speed-setting (fast or slow) it will keep your levels steady and gets very close to an engineer riding the faders manually. Make sure the output level stays at around -18dBFS - this goes for the output of all plugins in this chain.



The [Waves Vocal Rider](#) Plugin

2. SIBILANCE

Do you feel the sound of the vocal leans towards too much sibilance? Time to use a De-Esser, which is a specialized compressor that has an EQ in the sidechain. A De-Esser can reduce the level of syllables that have too much treble (typically between 5-12kHz). The frequency that you want to reduce can usually be set in the plugin, as well as the threshold (minimum level at which it starts working) and the amount of reduction in dB. You can use an analyzer to find out at which frequencies the "Esses" are jumping out.

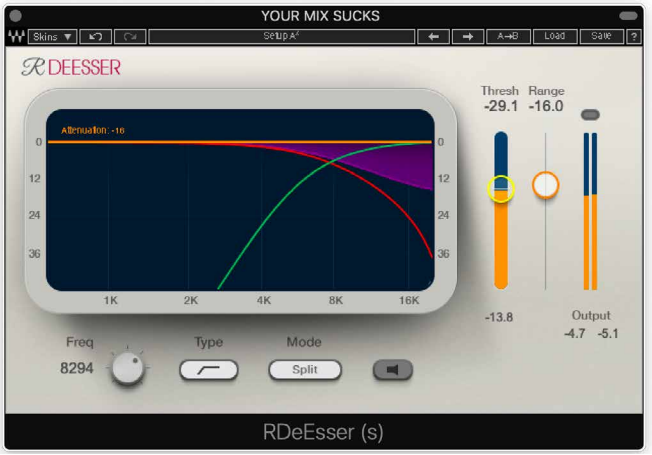
Alternatively, reducing the sibilance pre plugin chain is a more precise way to do this. In that case, I isolate the "Esses" to their own regions which I drag to a separate track but same channel strip (so I can select all "Esses" at the same time). I reduce the gain on these regions between 6dB - 10dB, depending on how much the vocal gets compressed in the mix. You want to set them low enough to run below the compressor/s threshold (If this is new to you, we'll talk about this in chapter 9).

If you can't get a De-Esser to achieve the same - use this technique! You can also completely isolate "Esses" to another channel strip and EQ/compress them differently. Same goes for breathing noises.

Waves has developed a total of 3 distinctly different De-Essers over the years, all of which can be useful depending on the situation. The latest Waves Sibilance plugin will cover you in all of these situations.

The original Waves DeEsser in "Wide" mode has been my "go to"-plugin for many years when I don't want the De-Esser to manipulate the sound. In "Wide"-mode, the De-Esser is just reducing the level, every time sibilance is occurring at the selected frequency. It's very quick and intuitive to dial in, and has a split mode as well which reduces sibilance at just the specific frequencies.

The Waves Renaissance DeEsser does a similar thing, but has a slightly different vibe to it - I like the way it smoothens out the sound of the "Esses" in "Split" Mode.



The Waves Renaissance DeEsser Plugin

The Waves Sibilance is their latest De-Esser, based on the Waves Organic ReSynthesis engine. It has the most variable and flexible settings including a cross-fader that can mix between wide and split-mode, very variable detection of sibilance and separate controls for threshold and range.



The Waves Sibilance Plugin

3. WARMTH

With the vocal sitting a little low in the track, we now need to develop a feeling for what's missing. We do that by adding frequencies using EQs, starting from the low mids.



The Waves PuigTec MEQ5 - The "Midrange Equalizer" is modeled after an original 1950s Pultec MEQ-5

A. Broad boost between 200Hz and 500Hz

The Waves PuigTec Midrange Equalizer MEQ 5 is usually my first EQ in the vocal chain, but you can simulate these (broad) curves with many other EQs. I don't ever go lower than 200Hz, and occasionally up to 700Hz.

The effect we want to get here is that the vocal gets more weight and warmth in the mix. If the vocal is well tracked, it comes with a lot of that quality in the recording and you may not need to do anything here.

This is why people use various transformer or tube-based equipment (from Mics to EQs/Compressors) during tracking.

However, a lot of modern vocal recordings sound rather thin, and a nice boost in the low mids can help that! If you like the character you're adding with the boost, you can even do a little bit too much of it.

We can then counterbalance that in the next steps. In case the vocal already sounds overly "muddy" or "boomy", add an EQ plugin like F6 RTA at the beginning of the plugin chain, locate and remove the frequencies that cause this effect. Watch the interdependence of that - once you've removed resonances, you have more leeway to use that broad Pultec-boost again.

You really can't go wrong with a Pultec, but there are of course other EQ emulations that do a similar job. We are looking at more EQ models in chapter 9.

B. Tube compressor for tone

After the boost, insert a compressor for tone - we are not after compression for level automation here, but want to add harmonics on top of the low mid boost and create a sense of glue. The classic Fairchild 660/670 (available by Waves as PuigChild Compressor) works great here, but don't limit yourself - many compressors can do the job. Just make sure the compression is very subtle.

A compressor after an EQ boost works so well because the two components interact. The more you boost a specific frequency, the more compression will be applied to that frequency. So the two balance each other out, which makes that combo very organic. Light tape saturation can also work here - and we are taking a closer look at more classic compressors in a later chapter!



The Waves PuigChild 660 is modeled after a rare Fairchild Compressor (The picture shows the mono version which is called 660, the stereo version is called 670)

4. PRESENCE

In a dense mix, this is were we create the frequencies that make the vocal cut through the rest of the instruments. We can go to extremes here, but before you start playing with the mid boost, set up the "attitude" compressor that follows it right away. It's needed to tame the mid boosts as they can get very harsh.

Often, less or no boosting in the mids is needed in less populated parts of the song, but when the vocals are up against a wall of sound, you will need a musically composed texture of "cut through" frequencies there.

Various boosts between 1kHz and 8kHz (SSL EQ)

This is more complicated to get right, compared to creating warmth. Start with a SSL-type EQ and boost the high shelf at 8kHz +10dB, then pull back again to 0dB and find a great setting for it somewhere in the middle. Try switching between bell and shelf characteristics (bell will just boost around the set frequency, while shelf also includes all frequencies above.) If 8k is boosting sibilance too much, go a tiny bit lower.

Continue by using the HMF band to boost at 4k. And the LMF to boost 2kHz. Move these around until you find a good balance - but keep in mind, not boosting anything is always an option. The goal is to create a cluster of mid-boosts that really becomes one colorful and musical texture of mid-boosts between 1kHz and 8kHz. You can achieve good results with stock EQs of your DAW, but there is a reason why SSL EQs are famous for their musicality in the mids. API works as well. Not a job for a Pultec.

5. ATTITUDE

This compressor's job is to tame the mid-boosts we've just created. We want to see it work hard and fast, but it needs to have a lot of musicality to reduce the harshness of the mid boosts.

The classic Bluestripe has become my favorite compressor for this stage and the harder you drive it, the more "attitude" you're getting. Waves has recently introduced the (Chris Lord-Alge, just sayin') CLA MixHub which, luckily, combines both a classic SSL-type EQ and the BlueStripe compressor into one plugin.



The Waves CLA MixHub

6. FINAL TONE AND DYNAMICS CONTROL

A. EQ

A final EQ can round off highs and mids. I like the Waves PuigTec EQP-1A here, to boost at 20Hz, and attenuate at 20kHz. This plugin is modeled after the legendary 1950s Pultec EQP-1 tube equalizer. Your vocals will sound more analog when you roll off the top end slightly.

Both the boost at 20Hz, and the attenuation at 20kHz should not affect the essence of the tone you have created. The boosts can add little bit of weight, and the cut removes top end energy that only hurts at loud volumes.



The [Waves PuigTec EQP-1A]

B. Brickwall Limiter

You can use a limiter at the end of the chain to catch occasional peaks that jump out. Only catch occasional peaks, and set the release time higher than 100ms. A digital brickwall limiter works well here. I happen to be a fan of the classic [Waves L1] in this place.



The [Waves L1 limiter] - a classic

PLEASE KEEP IN MIND!

The plugin chain outlined above is pretty complex. There is of course the possibility of processing the vocals too much with a signal chain like this. When you want a really natural vocal sound, you might just need 3B (the subtle tube compressor) and 6A (Pultec for basic tone control).

A natural vocal sound requires a very good recording though, and we rarely have an influence on that as mix engineers. On the other hand, this plugin chain is designed to be able to deal with any type of vocal, and when the vocal needs to cut through a dense mix, processing is absolutely necessary. However, subtle settings in each plugin can still cumulate in a drastic overall effect, while retaining a natural tone.

PARALLEL COMPRESSION

Chapter 8 is entirely dedicated to parallel processing, but it's worth mentioning here - if you master the vocal chain of this chapter, parallel compression will bring your vocal skills to another level:

On most songs that have a certain dynamic, and specifically ballads that start soft and end in a big showdown involving an entire band, orchestra, choir backing vocals, etc., I've made the following experience:

- The vocal chain for the (light) first verse, that has only a singer with a piano and (maybe) light drums, is found very quickly. A natural vocal sound utilizing 3B and 6A in our chain often works very well
- As the singer hits higher (and louder) notes, I need to add more elements of the vocal chain to control the dynamics, and also to tame certain resonances that come with loud and high notes
- On the chorus, especially towards the end, the vocal has to cut through a wall of sound, and several compressors are working in the chain
- The solution for the dilemma are two separate vocal channels: a natural one, and a processed one!
- Anything between the two is using a mix of those two channels - by automating both levels you can create the right blend between them for each section or even single notes
- Use this technique when the singer performs across a wide range of notes, switching into falsetto, shouting, hitting rough rock notes, etc.
- Just use a separate track for each of those styles, so you can vary settings and create a balance from these different tracks
- More details on this are coming in chapter 8

VOX AGAINST THE WORLD

I have worked on songs where the backing track was such a big and dense "wall of sound", that the vocal would not stick out of the mix. Multiband Sidechaining can be very useful here - plugins like the C6 Multiband Compressor with a sidechain for example:

- First check where the lead vocal has it's energy in the mids, this could be anywhere between 1kHz and 5kHz. Find the right frequency by sweeping through those frequencies with an EQ
- Make sure the backing track instruments and vocals are routed to two different busses, or create an extra bus or audio group for the instruments that make the arrangement dense in the mix, typically guitars, piano keyboard, orchestra sounds
- Insert the Multiband Compressor on that audio group - the one that has all of these mid range instruments and set it so that the side chained Multiband Compressor frees up the mids depending on the vocal's volume
- You can find the perfect setting where you don't notice it but the mids are pushed back just enough whenever the lead vocal needs that space!



The Waves C6 Multiband Compressor in sidechain-mode

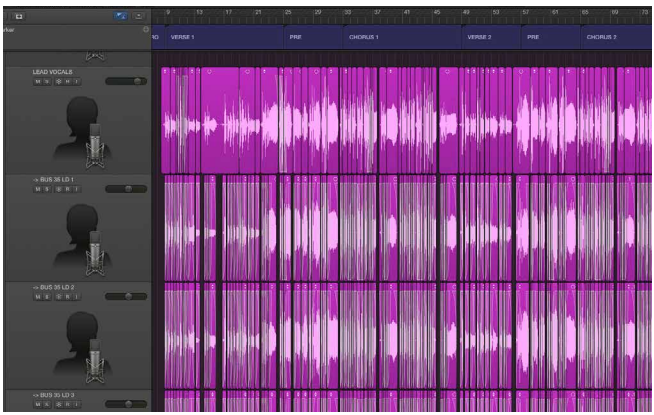
BACKING VOCALS

Backing Vocals are a lot easier to deal with compared to lead vocals - they usually come in groups of 3 and more, and they have to sit behind/underneath the lead vocals.

Whereas we strive to create a lot of dynamics, and preserve transients on lead vocals, the backing vocals can be more heavily treated. You can even merge several layers of backing vocals into one file for easier handling.

"All vocals please sing the same song,
at the same time!"

However, there is one thing that you can't be sloppy with and that's timing! I'm really sorry to reveal this to you now, but you have to zoom into the waveforms, and match the timing of the backing vocals and lead vocal doubles to the lead vocal!



Important details:
The backing and lead vocal timing must be spot on

While there's several software tools that promise to do this for you, you always have to double-check if you want a flawless result:

- "Esses" need to sit exactly in sync with where they sit in the lead vocal - otherwise those messy "Esses" will jump out of the mix left and right like the snakes on my favorite movie "Snakes on a Plane"; in addition to that, reduce them in level, they must not interfere with the sibilance of the lead vocal!
- Breathing noises in backing vocals? Pretty useless! You might enjoy the lead singer's breathing, but not 20 of them. Therefore, feel free to remove them



See this? I cut away the breathing noises in the backing vocals. Also, applying fades makes things even smoother

BACKING VOCAL CHAIN

As far as processing, backing vocals are a lot more forgiving. I usually copy and paste the lead vocal chain to the backing vocals as a starting point, then costumize on the following parameters:
(the numbers are referring to the plugin order of the lead vocal chain, which we use as a starting point)

1. Continuity
   Equally important for the backing vocals - treat same as lead vocals.
2. Sibilance
   Definitely use a DeEsser - backing vocals should have a lot less sibilances compared the lead vocals.
3. Warmth
   A. The Pultec-boost (MEQ)
   Boost at a different frequency relative to the lead vocal - boost. If the lead vocal has a "warmth"-boost at 300Hz, try 200Hz or 500Hz. If you have several backing vocals or LV (Lead Vocal) doubles, try to boost them at different points, so that the boosts are spread across a wide range.
   B. Tube compressor for tone
   Same as LV, but a bit more gain reduction.
4. Presence
   Here's where we go counter to the boosts of the lead vocal. Reduce the frequencies you have boosted on the LV.
5. Attitude = Bypass = don't use it!
6. Final Tone and Dynamics Control
   A. EQ: Adjust this while running all vocals. Attenuate the treble even more relative to the lead vocal. Try to boost at 12kHz or 16kHz to create some "shine" if applicable to the genre.
   B. Brickwall Limiter: You can limit a lot more on the backing vocal.

Question on filters and eq etc... The question is general but I'll use a high pass filter as an example. Going on older desks and outboard I've used, there was always some bleed that came through the filters, but with advancements and especially in the digital domain the cut is now sonically precise. I generally keep a little of the original there, especially with stems because that's what I've become used to.

To me it feels less clinical but also ends up a warmer tone. (or maybe I am imagining it) I even allow a tiny bit of bleed into the vocal tracks! So thinking ITB, and with all the tools there, is there any real danger of being too clinical and is there a better way of avoiding it?

What you're referring to as bleed is simply a less step filter (12dB/oct was sort of standard in analog); in general you did the right thing - however, the harder you compress the signal later on in the plugin chain, the more important it is to really get rid of the low rumble for example in a vocal, and in any recording made with a mic; you don't want this stuff to come back at you and start triggering the threshold of the compressor.

So, yes you could indeed end up with a less warm signal when you cut hard; I personally cut very hard with a 48db/oct linear phase eq plugin, and I go very high with the cut, to the the point where I cut into the lowest note of the vocal. I compensate for that with a Pultec tube EQ emulation as the first stage after that, and boost around the range of the fundamental note (between 200Hz - 500Hz depending on the range of the vocal) to bring that warmth back in a very controlled way.

When mixing a full album of varied dynamic material - what's the methodology of achieving a consistent vocal level throughout. Would you start the first mix using the more populated tracks as regards to instrumentation or the more open tracks with just stripped back instruments and use that vocal level for the rest of the album tracks?

Start with the more populated tracks, save plugin-chains of your lead vocal and start with a similar setting on the next song. Keep the songs you've already worked on at hand as a reference.

THE END

Thank you very much for taking the time to study this excerpt of the full e-book. This is actually not the end, as I am inviting you to join the online support group [YOURMIXSUCKS] on Facebook.

In this group you can ask specific questions about the book and I will continue to update the book as your questions are coming in. My goal is to create a complete software compendium on mixing that will stand the test of time. Assuming you're a licensed user ;-), you will receive lifelong updates of this book for free. And as I said in the beginning: I'd like to invite you to join me on Facebook, Instagram, YouTube and LinkedIn.
I am using all aforementioned channels as well as my podcasts (available in English and German language) to deliver a lot of free audio mixing and social-media marketing know-how for producers, DJs, singer/songwriters and audio engineers.

Also, all legitimate buyers of the full e-book receive a year of free mix feedback from me - you simply e-mail me your track to feedback@mixedbymarcmozart.com (please add this this e-mail-address to your contacts) and I'll come up with some quick suggestions for improvement. Of course, you can use that same e-mail for all sorts of feedback, questions and whatever you need from me.

Last but least: make sure to have a look at the mixedbymarcmozart.com shop. Apart from a lot of free mixing know-how, we have some quality offers going on from time to time.

Marc Mozart, November 2020